A Markov decision process for dynamic scheduling of recovery tasks during cascading failure

Sanfan Liu[0000−0002−5089−3500], Sahra Sedigh Sarvestani[0000−0002−2917−5643], and Ali R. Hurson[0000−0002−9510−0762]

Missouri University of Science and Technology
{slbdb,sedighs,hurson}@mst.edu

Abstract. Interdependencies among the components of a complex networked system make cascading failures more likely and more destructive. The research presented in this paper leverages knowledge of these interdependencies to identify components whose repair or replacement can lead to the quickest recovery of service capacity. Unlike related studies that focus on optimizing post-hazard recovery, the proposed approach acknowledges that failures can continue to propagate as recovery efforts are underway and considers the effect of failure propagation in scheduling recovery tasks. Two recovery schemes, with the respective goals of maximizing service recovery rate and minimizing cascading failure, are proposed and compared. Each uses a Markov decision process to dynamically prioritize components for recovery. We illustrate the recovery schemes by applying them to a smart grid based on the IEEE-14 test system. We compare the service capacity, recovery time, and number of cascading failures for the two dynamic schemes to randomized scheduling and a precomputed static scheme. The dynamic schemes, which are found to be superior in three different experiments, are especially useful in cyber-physical critical systems, where components are tightly coupled yet high availability is expected, so the system cannot be shut down for repair.

Keywords: Recovery model · Markov decision process · Failure propagation · Repair scheduling.

1 Introduction

Smart grids, which include renewable energy sources, wireless monitoring units, and intelligent control devices, serve as the prototypical example of a critical cyber-physical system expected to function reliably despite its complexity. Errors and malfunctions are inevitable, but their consequences can be severe, especially if they lead to cascading failures or blackouts.

Three notable examples of cascading failure recently occurred in Kenya, where three nationwide blackouts occurred between January 2022 and August 2023 [1]. The longest blackout in Kenya's history occurred on August 25, 2023, in a nationwide outage of over 14 hours that affected over 50 million people [1]. The triggering event for this blackout is believed to be an error message that caused the Lake Turkana Wind Power Plant to go offline [2], leading to a loss of 270 MW of power (approximately 15%) of total demand at the time. The loss triggered an imbalance in Kenya's power system and tripped all other main power generators, leading to total outage of the grid [3]. Efforts to restore power began as soon as power loss was detected in the affected line [3]. The first solution attempted was to cover the 270 MW power shortage by connecting to Uganda's neighboring power grid. This solution failed, for unknown reasons [2]. One of many consequences was the multi-hour stranding of hundreds of people at the country's main international airport, in Nairobi [2]. This example is one of many that highlight the importance of the careful selection of recovery actions and the serious consequences of cascading failures, which pose serious challenges to the stability, sustainability, and survivability of smart grids [4].

Mathematical modeling can prevent catastrophic outcomes by elucidating the effects of failure, predicting failure propagation paths, and enabling dynamic comparison of mitigation and recovery strategies. High-fidelity simulation of critical systems such as smart grids enables rapid, safe, and repeatable validation and evaluation of models.

To this end, this paper presents a Markov decision process (MDP) model that reflects the operation, failure, and recovery of a complex networked system. As described in Section 3.1, an MDP is a state-based model that compares different actions by associating a reward with each action. The model considers dependencies, including cyber-physical coupling, between components in predicting failure propagation. We quantify system-level functionality by the fraction of nominal service capacity delivered by the system. For example, system-level functionality of a smart grid can be quantified as the fraction of nominal power capacity successfully delivered by the system.

The fundamental distinction of our proposed approach is dynamic decision support that guides recovery efforts under circumstances where the initial disruption may lead to additional failures that require a change to the recovery strategy. This distinction increases the utility of our work for complex networked systems where tight coupling increases the chance of cascading failure, but the requirement for high availability limits or eliminates the possibility of isolating a failed area. Using service capacity as the measure of system functionality is a nod to survivability, an attribute of import for critical systems. We propose two dynamic recovery schemes, each of which considers failure propagation and recovery time in prioritizing components for recovery. One scheme favors resilience by prioritizing components whose recovery leads to the quickest restoration of service capacity. A second favors survivability by prioritizing the recovery of components whose failure is most likely to cause a cascade.

The proposed model enables comparisons of alternative recovery schemes by considering the availability of repair crews, predicting failure propagation, and estimating the level of service capacity over time. Given that the analysis is based on consequences rather than root causes of failure, the proposed approach can be applied to recovery from both accidental failure and deliberate attack.

2 Related Work

Two categories of related work are discussed in this section. The first category comprises studies on prediction, analysis, and mitigation of cascading failures, all of which are required for dynamic prioritization of recovery tasks in a tightly coupled system at risk of failure propagation. The second category is research focused on guiding the recovery of a complex system after failure.

2.1 Modeling of Cascading Failure

Research in this group aims to identify and quantify vulnerabilities and coupling effects within complex networked systems, with the goal of understanding the dynamic effects of disruption. A notable example is by X. Zhang et al. [5], who investigated the effect of cyber coupling on cascading failures in power systems, demonstrating that the coupling lowers the system's robustness and increases the extent and rate of failure propagation. Similarly, G. Zhang et al. [6] investigated coupling effects on the communication layer of a cyber-physical power system and analyzed the relationship between the coupling strength and the load loss rate on smart grids of different sizes. Xu and Ding [7] have proposed a model to investigate the effect of inter-network correlation on cascading failure in a cyber-physical system. They discovered that increasing interconnectivity can help enhance robustness against cascading failures in symmetrical networks, but has minimal effect in asymmetrical network systems. Nyberg and Westman [8] have created a framework to analyze failure propagation using contracts theory, with consideration of each component's requirements and specifications. They demonstrate that unmet requirements can increase the likelihood of, and in some cases, can guarantee the cascading failure.

These four studies, among others, show that tweaking the strength of coupling and/or enhancing the system with redundancy are key methods of preventing and mitigating the effects of cascading failures. Our proposed model goes further by reflecting and enabling analysis of the effects of recovery efforts undertaken concurrently with a cascading failure. We build on our earlier work [9], where we created a dependency index to represent the conditional probability of failure propagation between components of a cyber-physical system.

2.2 Modeling of the Recovery Process

Studies involving models for the recovery process typically focus on comparing recovery schemes, often by estimating the system's performance during recovery. The typical goal is to identify a scheme that enables the quickest recovery of system performance.

Notable work in this category includes studies by Yan et al. [10] and Sarkale et al. [11], where an MDP model has been applied to physical (not cyber-physical) power and water systems, respectively. The goal is to identify the dynamic repair scheduling algorithm with the highest recovery rate. In Yan's method, the extent of lost functionality is the basis for estimating the performance, and the scheme chosen is the one that recovers the most functionality. In contrast, the number of users who continue to receive service is the basis for comparison in Sarkale's method. Lin et al. [12] propose an MDP model for the maintenance of power supply equipment, where the goal is to minimize the sum of penalties and repair costs and reduce the cost of upkeep in the long run.

The aforementioned studies assume that recovery is undertaken when the system is immune to additional failures, because the system has been shut down for repair or isolation mechanisms are in place to prevent cascading failure. This is an unrealistic assumption for critical systems. The August 2023 blackout in Kenya, among other rolling blackouts, serves as a cautionary example.

In contrast, our proposed approach reflects the effects of failure propagation during recovery. Among the studies that address recovery from failure, our work is the only one that examines cyber-physical dependencies. Our measure of system functionality is the service capacity index (SCI), which is the fraction of full nominal service capacity delivered in a degraded state. Unlike the graph-theoretic metrics used in the majority of comparator studies, the SCI directly reflects the bottom line - service delivery. It can be defined for a variety of application domains and is not limited to power systems. Table 1 further highlights the distinction of our work by comparing it to the studies discussed earlier in this section.

Table 1. Comparison of proposed approach to related work

Reflects / Study	Component-level states	Physical layer	Cyber layer	Performance metric(s)	Failure propagation	Recovery process
This research	✓	✓	✓	service capacity index	✓	✓
Ref [5]	✓	✓	✓	number of nodes lost	✓	
Ref [6]	✓		✓	power load loss rate (%)	✓	
Ref [7]	✓	✓	✓	proportion of non-critical nodes (%)	✓	
Ref [10]	✓	✓		functionality loss (%)		✓
Ref [11]	✓	✓		number of users receiving service		✓
Ref [12]		✓		loss in power (MW) & repair cost($)		✓

3 Proposed Recovery Model

This section articulates our proposed model for the recovery process of a complex networked system comprised of individual components connected to enable delivery of a system-level service. The basis for our model is an MDP that enables comparison of recovery schemes that can differ in i) how components are prioritized for repair and ii) the number of crews available for recovery efforts.

The *components* are defined at a granularity that allows the state of each component - either *operational* or *failed* - to be directly observable. This is a reasonable assumption for systems with adequate diagnostics in place. A number of recent studies demonstrated the utility of wireless communication, sensors, and monitoring systems in detecting component-level failures in a cyber-physical system [13–15]. In a smart grid, the connected components can include physical elements such as power generators, buses, transmission lines, distribution lines, and communication lines, as well as cyber components for monitoring, decision support, and control.

Multiple components can fail or recover in each time step as failures occur and propagate or components are repaired. Failure propagation is non-deterministic and calculated from failure rates and component dependencies. All components are assumed to be recoverable, either through repair or replacement.

At this stage of the research, we assume that given adequate time and resources to repair or replace components, the system can recover from any failure. This assumption will be relaxed in our future work. We assume that historical data is available for each component's recovery time. Damage assessment reports from previous failure events can be used to this end. This is a reasonable and common assumption [10], given the abundance of diagnostic instrumentation and the documentation associated with repair of critical systems.

Resources required for recovery include repair crews who carry out the repairs or replacements. A known number of repair crews are assumed to be available during the recovery process. At this stage of our research, we assume that a given repair can be completed by any available crew; each crew is assumed to work on only one repair at a time. Given that multiple components can fail in a given simulation, multiple repair crews could be working concurrently. The ultimate goal of our research is to create a mathematical framework to enable decision support for the recovery process, i.e., to identify the best recovery strategy for a given failure and resource scenario.

3.1 Markov Decision Process

A Markov decision process (MDP) is a state-based model for sequential decision-making when actions can have uncertain outcomes. An MDP has four elements:

- A set of states, S
- A set of actions, A
- A set of rewards, R
- A set of transition probabilities, P

Probability $p_a(s, s') \in P$ denotes the probability that, at a given decision epoch, an action $a \in A$ will cause the system to transition from state $s \in S$ to state $s' \in S$ at the start of the next decision epoch, which begins when the action is complete and a reward of $R_a(s, s') \in R$ has been earned. An MDP is utilized to identify a sequence of actions that results in the highest expected reward.

In the proposed recovery model, the goal is to identify a sequence of recovery actions that maximizes a given objective, such as the system's service capacity upon completion of the recovery actions.

States The state occupied by the MDP at a given decision epoch is determined by the respective states of the components and the repair crews.

A given component, i, can be in one of two states at any given epoch: "operational" or "failed". The state of the component, x_i, is determined as in Equation 1.

$$x_i = \begin{cases} 1 & \text{if component } i \text{ has failed} \\ 0 & \text{if component } i \text{ is operational} \end{cases} \tag{1}$$

For a complex networked system with a total of N components, the system-level state is represented by the vector $\mathbf{x}$, as in Equation 2.

$$\mathbf{x} = \{x_1, x_2, \dots, x_N\} \tag{2}$$

Repair crews are the second aspect involved in identifying the state of the recovery process at a given decision epoch. Each repair crew, k, can be either "idle" or recovering a single component. The corresponding state, y_k, can be determined as in Equation 3, where $1 \leq i \leq N$ for an N-component system.

$$y_k = \begin{cases} 0 & \text{idle, no recovery task assigned} \\ i & \text{taking recovery actions on component } i \end{cases} \tag{3}$$

Given a total of M repair crews, the state of the crews is collectively represented by the vector $\mathbf{y}$, as in Equation 4.

$$\mathbf{y} = \{y_1, y_2, \dots, y_M\} \tag{4}$$

Actions: For a given decision epoch, feasible actions include all recovery actions that involve an available crew and a failed component that has not yet undergone recovery. This condition can be expressed as in Equation 5.

$$(\exists i \in [1, N] | x_i = 1) \wedge (\exists k \in [1, M] | y_k = 0) \tag{5}$$

Two crews cannot simultaneously recover the same component, i.e., for every $x_i \in \mathbf{x}$ where $x_i = 1$ there is at most one $y_k \in \mathbf{y}$ where $y_k = i$. Assignment of an action to a crew will cause a transition in the next time step from "idle" to a state that reflects the component being recovered by the crew. Scheduling of recovery actions is non-preemptive, i.e., a crew will remain on a given task through completion.

If action a takes the system from state $\mathbf{x}_\tau$ at time τ to state $\mathbf{x}_\gamma$ at time γ, the elements of the two vectors will be identical, except for elements that reflect the state of components that have become operational in state $\mathbf{x}_\gamma$ because their recovery was completed.

Rewards: The overarching goal assumed for the system is providing stable service, as quantified by the service capacity index (SCI), which reflects the fraction of nominal service capacity that is delivered at a given decision epoch. The value of the SCI depends on the system-level state, $\mathbf{x}$.

$$\mathrm{SCI}(\mathbf{x}_\tau) = \frac{\text{service capacity delivered in state } \mathbf{x}_\tau}{\text{nominal service capacity}} \tag{6}$$

In the proposed model, the reward for an action, a, is the change in the SCI.

$$R_a(\mathbf{x}_\tau, \mathbf{x}_\gamma) = SCI(\mathbf{x}_\gamma) - SCI(\mathbf{x}_\tau) \tag{7}$$

Transition Probabilities: A component can make transition from "operational" to "failed" due to independent failure or dependence on a failed component. The probability of the latter is quantified by the dependency index articulated in [9]. For a system with N components, the dependency indices can be represented by a matrix, D:

$$\begin{gathered} D = [d_{ij}] \in [0,1]^{N\times N}, \\ i \in [1,N], j \in [1,N] \end{gathered} \tag{8}$$

d_{ij} represents the conditional probability of component j failing one time step after component i has failed. The heuristic causation analysis we used to compute the value of each element in D is described in detail in our earlier work [9]. Other methods, e.g., the interaction model proposed by Qi et al. [16] can also be used to populate D.

Given d_{ij}, the probability that component j will fail in the next time step in a cascade from i is:

$$p_{ij} = p(x_j(t+1) = 1 | x_i(t) = 1) = x_i \cdot d_{ij} \tag{9}$$

To find the total cascading effects on component j, a summation from every component in the system must be calculated:

$$c_j = p(x_j(t+1) = 1 | x_j(t) = 0) = \sum_{i=1}^{N} p_{ij} \tag{10}$$

A transition from "failed" to "operational" for a component results from recovery, i.e., repair or replacement by a crew.

The estimated recovery time (in steps) for a given component, i, can be learned from historical data and represented by the probability distribution of repair time T_{ri}. Instant recovery is assumed to be infeasible.

The state of a repair crew transitions upon assignment or completion of a recovery task.

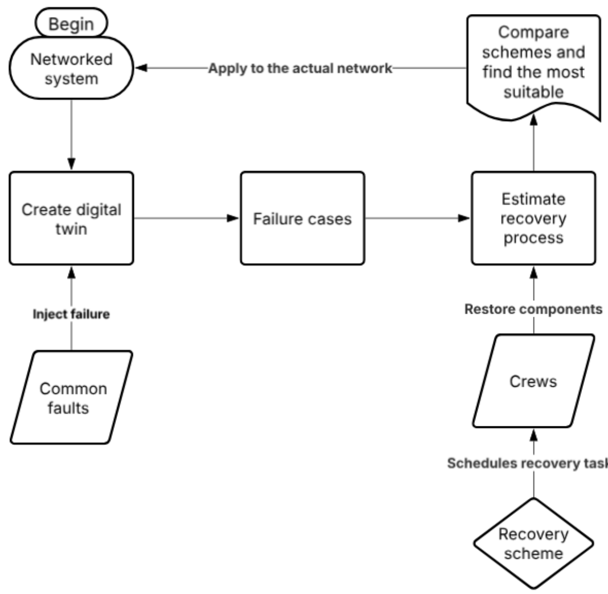

Fig. 1. Framework for comparing recovery schemes

3.2 Recovery Schemes

We demonstrate the recovery model by applying it to the four recovery schemes described below, which differ in how they prioritize components for recovery.

1. The *Randomized* scheme randomly selects a failed component at each decision epoch and assigns it to an available crew for recovery.
2. The *Precomputed* scheme uses (static) priority ordering that is computed beforehand. The priorities could be determined using component importance analysis [17], expert opinion, or other methods. At each decision epoch, this scheme schedules recovery for the failed component that has the highest priority on the precomputed list.
3. The *Max-capacity* scheme is a dynamic prioritization scheme we adapted from the method proposed by Yan at el. [10]. Our scheme aims to increase service resilience by prioritizing the repair of components that achieve the greatest recovery rate for service capacity. Consider two system-level states, $\mathbf{x}_\tau$ and $\mathbf{x}_\gamma$, which reflect the same state for every component other than i. $\mathbf{x}_\tau$ reflects the failure of component i, i.e., $x_i = 1$. $\mathbf{x}_\gamma$ is identical to $\mathbf{x}_\tau$ in every other element, but $x_i = 0$. Let action a represent the recovery of component i in between these two states. The reward for this action represents the marginal effect of recovering component i and is computed as the recovery rate of service capacity between these two states, as shown in Equation 11. In this equation, $E[T_{ri}]$ represents the expected repair time of component i, which is computed from historical data.

$$\text{Marginal effect of recovering } i = \frac{R_a(\mathbf{x}_\tau, \mathbf{x}_\gamma)}{E[T_{ri}]} \quad (11)$$

4. The *Min-cascade* scheme is a dynamic prioritization scheme we created for this research. This scheme aims to decrease the likelihood of failure propagation by prioritizing the repair of components most *critical* to the system. In this context, a component's criticality is a measure of the number of other components it will bring down if it fails. This scheme considers two elements in calculating a *cascading index* for each failed component, i. The first is the expected number of currently operational components that will fail in the next time step as a result of i's failure. This is the numerator of the fraction in Equation 12. The second is i's expected recovery time, $E[T_{ri}]$, which is the denominator of the fraction in Equation 12. The component with the highest cascading index is selected for recovery.

$$\text{cascading index of } i = \frac{\sum_{j=1}^{N} d_{ij} * (1 - x_j)}{E[T_{ri}]} \quad (12)$$

Table 2. Summary of recovery schemes

Schemes	**Order of recovery tasks**
Randomized	Random
Precomputed	Computed beforehand
Max-capacity [10]	Aims to maximize recovered capacity
Min-cascade	Aims to minimize failure propagation

Table 2 provides a summary of the four recovery schemes. These schemes differ in computational complexity. The randomized scheme requires minimal computational resources compared to the others, as it randomly chooses a repair task for the next idle crew. In our experiments, the Pre-Computed scheme was on average 10% slower than Randomized. This is not surprising, given that the former requires searching a list. The two dynamic schemes, Max-capacity and Min-cascade, are considerably more complex, as they look ahead in time and predict failure propagation. In our experiments (described in Section 4, Min-Cascade was on average 20% slower than the Randomized scheme. Max-Capacity requires performance evaluation of the network; its computational complexity can be reduced by using a precomputed lookup table to determine the service capacity for each system-level state. We found it to be on average 60% slower than Randomized despite our use of the table.

4 Case Studies

To demonstrate that our recovery process model can correctly differentiate between the recovery rate of different schemes, we present the results of three case studies. In the interest of brevity and clarity, all three cases assume the same underlying cyber-physical system structure - a smart grid based on the commonly used IEEE 14-bus test system (IEEE-14), depicted in Figure 2.

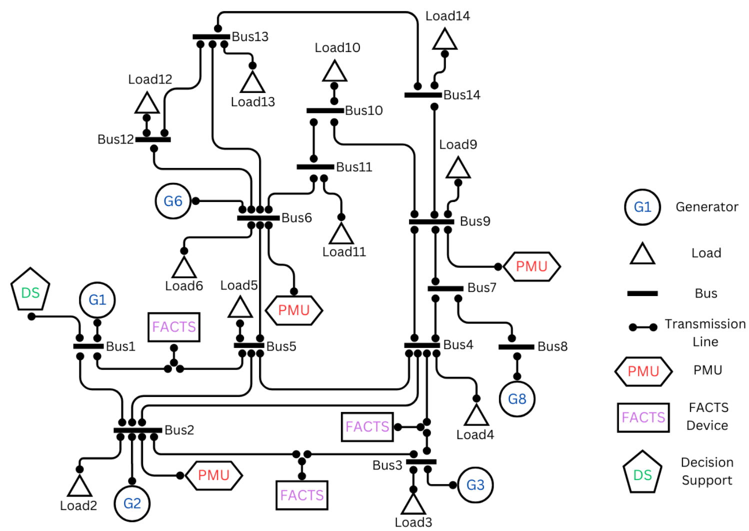

Fig. 2. IEEE-14 smart grid (adapted from [9])

The original IEEE-14 includes five generators, twenty transmission lines, fourteen buses, and eleven loads. We have created an IEEE-14 smart grid from the original physical system by adding three phasor measurement units (PMUs), three Flexible Alternating Current Transmission System (FACTS) devices, and a decision support system. The PMUs, which are high-speed sensors that carry out time-synchronized monitoring of voltages and currents on a power grid, were placed on buses 2, 6, and 9, respectively. These are the locations identified in a study by Asprou and Kyriakides [18] as leading to the most accurate estimates of the state of the grid. FACTS devices are used to control power flow and increase stability in power grids. They are widely used in the construction of smart grids [19] and offer advantages over mechanical switching devices [20]. We added three FACTS devices to the IEEE-14 grid, on the lines between buses 1 and 5, 2 and 3, 3 and 4, respectively. The locations were recommended by Acharya and Mithulananthan [21] to achieve the highest stability in voltage and power factor. Finally, the decision support system was added to bus 1, which connects to the generator that typically provides the greatest fraction of the power. The role of the decision support system is to determine optimal settings for the FACTS device based on data from the PMUs. Among the physical components of the IEEE-14 smart grid, the focus of our analysis is on transmission lines, as they have higher failure rates as compared to other components and are often the cause of power outages [22]. The failure of cyber components, i.e., PMUs, FACTS devices, and the decision support system, is also considered, as their malfunction can lead to cascading failure in an otherwise stable grid [23].

A total of 27 components are considered for recovery, hence $N = 27$. The system-level state, $\mathbf{x}$, is represented as in Equation 13, where $\mathbf{x_i} = 1$ if component i has failed.

$$\mathbf{x} = \{x_1, x_2, \ldots, x_{27}\} \tag{13}$$

For the given IEEE-14 smart grid, the recovery time for each component is assumed to be uniformly distributed, ranging from 2 to 4 hours for a transmission line and 3 to 7 hours for the smart devices. These values have been assumed for illustrative purposes. Actual recovery times depend on the type of fault and can be inferred from historical data [24].

4.1 Simulation Environment

The goal of the case studies is to show that the recovery scheme identified by the MDP can improve the delivered service capacity as compared to randomized assignment of recovery tasks, or prioritization that neglects the failure propagation that can be concurrent with recovery efforts. Our experiments were carried out using fault injection on PSAT [25], an open-source MATLAB-based toolbox for analysis of power systems, which we have modified to allow for simulation of interactions between components such as PMUs and the decision support algorithm that identifies settings for the FACTS devices. Our previous work [9, 26] describes the simulation environment and explains how we ensured that cyber-physical interdependencies are accurately reflected in our experiments.

For each case study, our simulator requires three inputs: i) a description of the topology and specifications of the smart grid (including the recovery time distribution for each component) and ii) the number of crews available to restore the system. From this information, the simulator carries out power flow analysis to determine the active power flow on each line and the voltage at each bus. The phasor data is computed by the PMUs from this information and sent to the decision support system, which determines settings for the FACTS device, which in turn may regulate active and reactive power on the line where it is installed. A second round of power flow analysis by the simulator determines the result of this regulation and identifies overloaded lines and failed cyber components, reflecting failure propagation in the smart grid. A failed component remains failed until a repair crew completes its recovery.

The MDP was formulated and solved in Python 3 on Google colab [27], with three inputs: i) the dependency matrix (described in Section 3.1) that governs transition probabilities and reflects cascading failure, ii) the recovery time distribution for each component, and iii) a look-up table that shows the service capacity, SCI_τ (defined in Equation 6) for each system-level state, $\mathbf{x}_\tau$. In our case studies, the SCI is determined by the fraction of loads that are connected to buses operating within the expected voltage range. The expected range is 0.9 to 1.1 per-unit, where the per-unit representation denotes normalization by the base voltage of each bus. Pre-computing the SCI values increases the efficiency of the simulation.

4.2 IEEE-14 Failure Cases

The assumptions underlying our model are described in Section 3. One of them is that component states are observable, i.e., the failure of a component will be detected and repair efforts initiated quickly. Given that single-failure contingencies are the most probable, in a short-sighted approach, repair crews are likely to be dispatched to the first component whose failure is detected. However, given that an initial failure can propagate to other components, prediction of failure sequences could lead to more prudent dispatch of repair crews.

Among the components in the smart grid, transmission lines, PMUs, FACTS devices, and the decision support system are the ones most likely to fail [22]. The respective failure of each of these 27 components is the triggering event for one of the 27 failure cases analyzed in our case study. For each failure case, we ran 2,000 Monte Carlo simulation trials with a respective recovery time horizon of 100 hours and compared the number of cascading failures, time to first recovery, and service capacity achieved by each of the four recovery schemes summarized in Table 2. The *time to first recovery* refers to the duration between injection of the failure and when all failed components have been repaired. The system may experience additional failures after this first recovery. Three separate experiments were conducted, with three, four, and five recovery crews, respectively.

Experiment 1: Three recovery crews available

In the first experiment, we compared the four recovery schemes as respectively carried out by a total of three recovery crews. Failures can propagate as repair or replacement is being carried out, leading to cascading failures that could have been prevented by faster recovery efforts. In this experiment, as expected, the recovery scheme designed to reduce cascading failure outperforms the others. Table 3 summarizes the results of the experiment.

Table 3. Comparison of recovery schemes, three crews

Scheme	Avg. time to first recovery (h)	Avg. number of cascading failures
Randomized	20.4	14.4
Precomputed	17.4	14.1
Max-capacity	19.6	14.7
Min-cascade	17.7	12.7

Figures 3 and 4 compare the service capacity achieved by the four recovery schemes. Min-cascade and Precomputed were able to reduce the number of cascading failures and the average time it takes to restore the entire system.

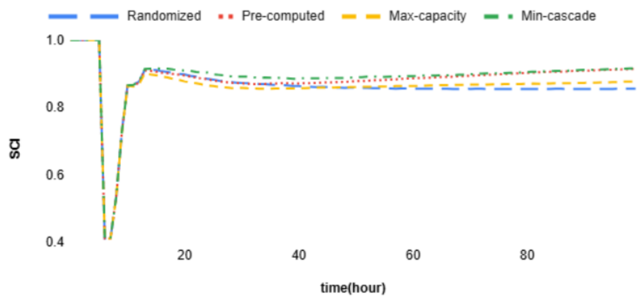

Fig. 3. Comparison of service capacity, three crews

The Randomized scheme performs poorly by both metrics. The greedy approach taken by Max-capacity backfires, as recovery cannot keep up with failure propagation. The Min-cascade scheme maintained the highest service capacity for the largest fraction of the time horizon. Reducing the chance of cascading failure buys more time to request additional recovery resources.

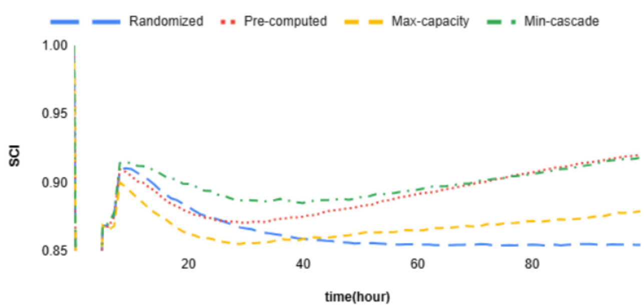

Fig. 4. Focused view of service capacity during recovery, three crews

Experiment 2: Four recovery crews available
In the second experiment, four crews were assumed to be available. Table 4 summarizes the results of the experiment. As expected, both the time to first recovery and the number of cascading failures decrease when more hands are on deck. Figure 5 compares the service capacity achieved by each of the four schemes during the recovery process. Overall, the Min-cascade scheme shows the greatest efficacy. In comparison to experiment 1, the SCI is more stable and less degradation is experienced by users, especially in the Randomized and Max-capacity schemes, respectively.
Experiment 3: Five recovery crews available
In the third experiment, we assume abundant recovery resources, i.e., the availability of five recovery crews. Given the size of the IEEE-14, having five crews

Table 4. Comparison of recovery schemes, four crews

Scheme	Avg. time to first recovery (h)	Avg. number of cascading failures
Randomized	14.1	13.0
Precomputed	14.2	14.8
Max-capacity	14.5	12.3
Min-cascade	13.3	11.0

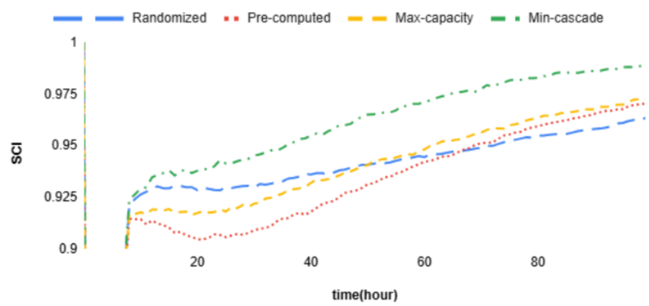

Fig. 5. Comparison of service capacity, four crews

available for recovery is admittedly unlikely to occur in reality, but useful in comparing how the number of crews affects recovery. Table 5 summarizes the results of the experiment. The results indicate that the Max-capacity and Min-cascade schemes can achieve the quickest recovery time and the least cascading failures. Their time to first recovery is similar. This is not surprising, given that both of them dynamically adapt to the consequences of failure and recovery. Figure 6 compares the service capacity achieved by the four recovery schemes.

Table 5. Comparison of recovery schemes, five crews available

Scheme	Avg. time to first recovery (h)	Avg. number of cascading failures
Randomized	12.2	11.5
Precomputed	13.4	16.0
Max-capacity	11.9	11.0
Min-cascade	11.8	11.0

We observe that the initial failure can quickly bring down more than three lines of the system, leaving it defunct for a very short time. The abundance of repair crews allows for rapid recovery.

Tables 6 and 7 summarize the results of the 12 experiments, respectively comparing the average time to first recovery and average number of cascading failures for the four recovery schemes for different numbers of available crews. Overall, the Min-cascade scheme is found to be superior, achieving low time to first recovery and the lowest number of cascading failures.

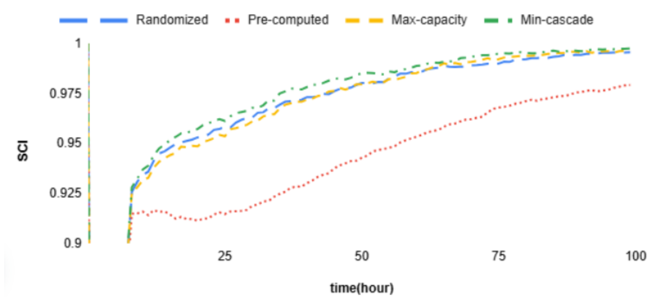

Fig. 6. Comparison of service capacity, five crews

Table 6. Avg. time to first recovery for each scheme

Crews	Randomized	Precomputed	Max-capacity	Min-cascade
3	20.4	17.4	19.6	17.7
4	14.1	14.2	14.5	13.3
5	12.2	13.4	11.9	11.8

Table 7. Avg. number of cascading failures for each scheme

crews	Randomized	Precomputed	Max-capacity	Min-cascade
3	14.4	14.1	14.7	12.7
4	13.0	14.8	12.3	11.0
5	11.5	16.0	11.0	11.0

5 Conclusions

This paper proposes a Markov decision process model for analysis and optimization of the recovery process of a complex networked system. We also propose two dynamic recovery schemes that predict and adapt recovery efforts to the consequences of failure propagation. The first scheme, adapted from [10], aims to achieve the quickest possible recovery of service capacity. The second scheme aims to minimize cascading failure. In a case study on the IEEE-14 smart grid, we compared the efficacy of the dynamic schemes to a randomized and a static scheme, respectively, for three scenarios that differ in the availability of repair resources (recovery crews). The results showed that each dynamic scheme achieved the objective for which it was designed. The Min-cascade scheme was the best overall, in terms of both time to first recovery and number of cascading failures.

The immediate future work of this research is to validate scalability of the model by applying it to larger smart grids, e.g., IEEE-34 and 57, as well as complex networked systems from other application domains. Other future extensions include the following:

Relaxing the assumption of observability of states Considering the effect of incomplete or uncertain information about the state of each component would be a closer representation of reality.

Considering additional recovery schemes We compared four schemes in this study. Several others have been demonstrated in respective case studies by Yan et al. [10] and Sarkale et al. [11]. Insights can be gained by comparing their efficacy.

Considering other performance metrics Other than the loss in service capacity, there are many other ways to evaluate the performance of a complex networked system. Domain-specific examples include the stability of bus voltages and the cost of power production/distribution/storage. These metrics could be used as the basis for comparing the efficacy of recovery schemes. A multidimensional metric could yield insight about tradeoffs.

Packaging and prototyping A user-friendly interface to the model would make its real-time decision support accessible to system operators.

6 Acknowledgements

The authors gratefully acknowledge funding from NSF DUE-2221559 and DEd GAANN award P200A210121. We extend our warmest thanks to Mr. John Czaicki, former Vice President of Quanta Utility Engineering Services, for sharing his professional insights.

References

1. The Associated Press, "Much of Kenya falls into darkness in the third nationwide power blackout in 3 months," *The Associated Press*, Dec. 12 2023. [Online]. Available: https://apnews.com/article/kenya-power-cut-blackout-nairobi-kenyatta-airport-d694c634fa84f438f2e9285ad685e3a3
2. Anna, Cara, "Cause of Kenya's longest power outage in memory remains unclear as grid suppliers exchange blame," *The Associated Press*, Aug. 27 2023. [Online]. Available: https://apnews.com/article/kenya-electricity-outage-politics-a681463711c756d415619c2e8743fd14
3. D. Mwangi, "Kenya Power: Why total restoration of power is taking longer than expected," *Pulse Live Kenya*, Aug. 26 2023. [Online]. Available: https://www.pulselive.co.ke/business/domestic/kenya-power-reveals-cause-of-nationwide-power-blackout-and-delays-in-restoration/28dyfmp
4. Y. Xiaohui, Z. Wuzhi, S. Xinli, W. Guoyang, L. Tao, and S. Zhida, "Review on power system cascading failure thoeries and studies," in *Proc Intl Conf on Probabilistic Methods Applied to Power Systems (PMAPS)*, 2016, pp. 1–6.
5. X. Zhang, D. Liu, C. Zhan, and C. K. Tse, "Effects of cyber coupling on cascading failures in power systems," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 7, no. 2, pp. 228–238, 2017.
6. G. Zhang, J. Shi, S. Huang, J. Wang, and H. Jiang, "A cascading failure model considering operation characteristics of the communication layer," *IEEE Access*, vol. 9, pp. 9493–9504, 2021.
7. Z. Xu and L. Ding, "Cascading robustness analysis of cyber-physical systems considering inter-network degree correlation," in *Proc 5th Intl Seminar on Artificial Intelligence, Networking and Information Technology (AINIT)*, 2024, pp. 457–460.
8. M. Nyberg and J. Westman, "Failure propagation modeling based on contracts theory," in *Proc 11th European Dependable Computing Conf (EDCC)*, 2015, pp. 108–119.
9. K. Marashi, S. Sedigh Sarvestani, and A. R. Hurson, "Identification of interdependencies and prediction of fault propagation for cyber–physical systems," *Reliability Engineering & System Safety*, vol. 215, p. 107787, 2021.
10. J. Yan, B. Hu, K. Xie, T. Niu, C. Li, and H.-M. Tai, "Dynamic repair scheduling for transmission systems based on look-ahead strategy approximation," *IEEE Transactions on Power Systems*, vol. 36, no. 4, pp. 2918–2933, 2021.
11. Y. Sarkale, S. Nozhati, E. K. P. Chong, B. R. Ellingwood, and H. Mahmoud, "Solving Markov decision processes for network-level post-hazard recovery via simulation optimization and rollout," in *Proc IEEE 14th Intl Conf on Automation Science and Engineering (CASE)*, 2018, pp. 906–912.
12. S. Lin, R. Fan, D. Feng, C. Yang, Q. Wang, and S. Gao, "Condition-based maintenance for traction power supply equipment based on partially observable Markov decision process," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 1, pp. 175–189, 2022.
13. S. Aust, R. V. Prasad, and I. G. Niemegeers, "Performance evaluation of sub 1 GHz wireless sensor networks for the smart grid," in *Proc 37th Annual IEEE Conf on Local Computer Networks*, 2012, pp. 292–295.
14. S. Verma and P. Rana, "Wireless communication application in smart grid: An overview," in *Proc Innovative Applications of Computational Intelligence on Power, Energy and Controls with their Impact on Humanity (CIPECH)*, 2014, pp. 310–314.

15. F. Tashtarian, A. Montazerolghaem, and M. Abbasi, "Distributed lifetime optimization of wireless sensor networks in smart grid," in *Proc IEEE Intl Conf on Smart Energy Grid Engineering (SEGE)*, 2018, pp. 240–243, ISSN: 2575-2693.
16. J. Qi, K. Sun, and S. Mei, "An interaction model for simulation and mitigation of cascading failures," *IEEE Transactions on Power Systems*, vol. 30, no. 2, pp. 804–819, 2015.
17. I. Alobaidi, S. Sedigh Sarvestani, and A. R. Hurson, "Survivability analysis and recovery support for smart grids," in *Proc of the 4th Intl Symp on Resilient Cyber Systems*, Chicago, USA, Aug. 2016, pp. 33–39.
18. M. Asprou and E. Kyriakides, "Optimal PMU placement for improving hybrid state estimator accuracy," in *IEEE Trondheim PowerTech*, 2011, pp. 1–7. [Online]. Available: https://ieeexplore.ieee.org/document/6019247
19. E. Sosnina, R. Bedretdinov, and A. Ivanov, "Assessment of FACTS devices non-sinusoidality in smart grid," in *Proc 20th Intl Conf on Harmonics & Quality of Power (ICHQP)*, 2022, pp. 1–5.
20. V. M. Garrido, O. D. Montoya, L. Alba, and D. Jimenez, "Location of FACTS devices in power systems: Application to the IEEE 9 bus system," in *Proc 26th IEEE Intl Conf on Electronics, Electrical Engineering and Computing (INTERCON)*, 2019, pp. 1–4.
21. N. Acharya and N. Mithulananthan, "Locating series FACTS devices for congestion management in deregulated electricity markets," *Electric Power Systems Research*, vol. 77, no. 3, pp. 352–360, 2007. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0378779606000812
22. Y. Song and B. Wang, "Survey on reliability of power electronic systems," *IEEE Transactions on Power Electronics*, vol. 28, no. 1, pp. 591–604, 2013.
23. A. Faza, S. Sedigh, and B. McMillin, "Integrated cyber-physical fault injection for reliability analysis of the smart grid," in *Proc 29th Intl Conf on Computer Safety, Reliability and Security (SAFECOMP 2010)*, Vienna, Austria, Sept. 2010, pp. 277–290.
24. K. K. Vadivel, "Emergency restoration of high voltage transmission lines," *The Open Civil Engineering Journal*, vol. 11, 2017.
25. F. Milano, "An open source power system analysis toolbox," *IEEE Transactions on Power Systems*, vol. 20, no. 3, pp. 1199–1206, August 2005.
26. K. Marashi, S. Sedigh Sarvestani, and A. R. Hurson, "Consideration of cyber-physical interdependencies in reliability modeling of smart grids," *IEEE Transactions on Sustainable Computing*, vol. 3, no. 2, pp. 73 – 83, April - June 2018.
27. "Google Colaboratory (colab)," https://research.google.com/colaboratory/faq.html, accessed: 2025-04-15.